Thanisorn Jarudilokkul

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Work Experience

AI Researcher Intern, Metallurgy Laboratory **Jul 2026 – Present**
CLP Power Hong Kong Limited (CLP Group), Hong Kong SAR

– Built a computer-vision pipeline combining YOLO detection, CNN classification, and logistic regression to automate defect grading of power-plant boiler tubes, reducing inspection time by over 65% and achieving 75% accuracy.

AI & Robotics Engineering Intern, Applied ML **Jun 2025 – Aug 2025**
Hong Kong Productivity Council, Hong Kong SAR

– Designed a multimodal digital-human system with neural voice synthesis and GAN-based facial animation, integrated with the DeepSeek API to generate dialogue based on specific information and roles.
– Worked directly with the marketing department to scope product requirements for the digital-human project.

Research Assistant, Hardware & System Prototyping **Jun 2024 – Jul 2024**
Institute of Field Robotics (FIBO), KMUTT, Bangkok, Thailand

– Engineered a low-latency embedded control loop in C++, implementing inverse kinematics via Jacobian pseudo-inverse with encoder-feedback correction on a 100 ms cycle, holding 95% positional accuracy on a 3D-printed robotic arm.

Education

The Chinese University of Hong Kong **Aug 2024 – Jun 2028 (anticipated)**
BEng, Mechanical & Automation Engineering (Robotics Stream), Minor in Computer Science **CGPA: 3.643/4.0**

– Related Coursework: Complex Variables (A), Differential Equations (A), Linear Algebra (A-), Multivariable Calculus (A-).
– Scholarships: HKSAR Government Scholarship, APEC Scholarship, Hang Seng Bank Scholarship, Faculty of Engineering Admission Scholarship.

Selected Projects

LOBExecutionEnv: Hierarchical RL for Limit Order Book Execution **Aug 2026**
– Built a FIFO limit order book simulator environment using Bitcoin L2 data and trained a hierarchical multi-agent system, including a Qwen LLM macro analyst, a SAC strategist, and a RecurrentPPO execution agent.
– Cut training time 2.5× by fixing a JSON-decoding bottleneck and vectorizing across parallel environments.
– Benchmarked against TWAP: implementation shortfall within 0.1 bps (0.99 vs. 0.89 bps) while agreement with its supervised classifier fell from 99.7% to 78.3%, showing the agent learned its own policy.

GVHMR2PBHC: Physics-Based Whole-Body Control for a Humanoid Robot **Aug 2025**
– Built a computer-vision motion-retargeting pipeline, converting human motion from video into robot joint trajectories for the Unitree G1 humanoid robot; achieved over 91% motion fidelity in real-world demonstrations.
– Trained a whole-body control policy using PyTorch and GPU-accelerated reinforcement learning in Isaac Gym and MuJoCo.
– Achieved adoption by robotics researchers at BIT, Shanghai University, ECNU, and USTB.

Technical Skills

Programming: Python, C, C++, HTML, CSS, JavaScript, SQL, MATLAB
AI/ML: PyTorch, TensorFlow, Transformers, Reinforcement Learning (RecurrentPPO, SAC), Computer Vision (YOLO, ResNet), CUDA
Quant/Data: NumPy, pandas, PyArrow, Statistical Testing, Market Microstructure, Implementation Shortfall, TWAP Benchmarking
Systems & Tools: Linux (Ubuntu), Docker, Azure, Git, REST APIs, Arduino IDE, Isaac Gym

Awards & Achievements

Goldman Sachs Engineering Academy, Hong Kong **Sep 2026**
Dean's List, Faculty of Engineering — Term GPA 3.94 **2025 – 2026**
Head's List, United College **2024 – 2025**
Honorable Mention (Team) & Special Diploma (Individual), 6th Mathematical Modelling Tournament, Moscow State University **Nov 2023**
3rd Place, e-ICON World Contest — VERITAS, AI-enabled mobile app for SDG4 **Aug 2023**
Honorable Mention, 9th International Mathematical Modeling Challenge (IMMC) **May 2023**

Leadership

Head of Activities, Thai Students Association, CUHK **Feb 2025 – Jan 2026**
Led the planning and execution of a Thai dessert workshop, coordinating and overseeing a team responsible for registration, logistics, and university and guest relations for 80+ attendees.